

Matching Polytope and Dual

For a weight vector $w = \{w_e\}$ on the edges
the max weight matching problem is:

$$\left[\begin{array}{l} \max: \sum w_e x_e \\ \text{st: } x(\delta(v)) \leq 1 \quad y_v \\ x(E[u]) \leq \lfloor \frac{1}{2} |u| \rfloor \quad \forall u \in V, |u| \geq 3 \quad \text{and } z_u \\ x \geq 0 \end{array} \right.$$

$$\left[\begin{array}{l} \min: \sum_{v \in V} y_v + \sum_{|u| \geq 3} z_u \cdot \lfloor \frac{1}{2} |u| \rfloor \\ \text{st: } \sum_{\substack{v \in V: \\ e \in \delta(v)}} y_v + \sum_{\substack{|u| \geq 3 \\ e \in E[u]}} z_u \geq w_e \quad \forall e \\ y, z \geq 0 \end{array} \right.$$

The Cunningham - Marsh Formula

Thm: If the weight vector w is integral, then:

a. y, z opt dual solution can be taken to be integral.

b. Moreover, can take z such that

$$\Omega = \{u : z_u > 0\} = \text{supp}(z)$$

is laminar. Ω is laminar if $u_1, u_2 \in \Omega$

$$u_1 \subseteq u_2 \text{ or } u_2 \subseteq u_1 \text{ or } u_1 \cap u_2 = \emptyset.$$

The proof — 1/4

Induction on $|E| + \sum_{e \in E} w_e$

Thus we can assume that $w_e \geq 1, \forall e$.

Exercise: show that we can apply

inductive hypothesis if $\exists$ vertex $\hat{v} \in V$ that is
present in every max weight matching.

The proof — 2/4

Therefore have left to consider graphs G
and weight vectors w s.t. $\forall v \in V, \exists \text{ max}$
weight matching M that misses v .

Claim: This implies $y_v = 0 \forall v$, i.e. $y \equiv 0$
in any optimal solution.

Follows from Comp. Slackness: $y_v (1 - x(\delta(v))) = 0 \forall v$

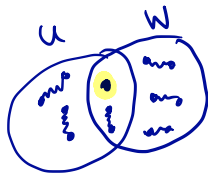
Next: choose opt. dual solution z maximizing: $\sum z_u \cdot \lfloor \frac{1}{2} |u| \rfloor^2$
and let $\Omega = \text{supp}(z) = \{u : |u| \geq 3, \text{ odd}, z_u > 0\}$.

Claim: Ω is laminar.

The proof — 3/4

Suppose claim false. Then $\exists u, w \in \Omega$
w/ $u \cap w = \emptyset$ and $u \not\subseteq w \not\subseteq u$.

$$z_u, z_w > 0 \Rightarrow \left. \begin{aligned} x(E[u]) &= \lfloor \frac{1}{2}|u| \rfloor \\ x(E[w]) &= \lfloor \frac{1}{2}|w| \rfloor \end{aligned} \right\} \text{CS} \quad *$$



$\hat{v} \in U \cap W$. Let M be
a max wt matching misses $\hat{v}$.
Let x correspond to M .

$$\Rightarrow |U \cap W| \text{ odd.}$$

$$\text{Let } \alpha = \min \{z_u, z_w\} > 0$$

Decrease z_u, z_w by α and increase $z_{u \cap w}, z_{u \cup w}$ by α .

Check: still feasible. Still optimal b/c obj not changed. But
we increased $\sum z_u \cdot \lfloor \frac{1}{2}|u| \rfloor^2$. Proves claim: Ω laminar.

The proof — 4/4



Suppose z not integral.

Take inclusion-wise maximal $\hat{U} \in \Omega$

with $z_{\hat{U}}$ non-integral.

Let $U_1, \dots, U_k \subseteq \hat{U}$ inclusion-wise maximal subsets of $\hat{U}$. (possibly $k=0$).

$$\text{Let } \alpha = z_{\hat{U}} - \lfloor z_{\hat{U}} \rfloor > 0$$

$$\text{New solution: } \tilde{z}_{\hat{U}} \leftarrow z_{\hat{U}} - \alpha, \quad \tilde{z}_{U_i} \leftarrow z_{U_i} + \alpha.$$

$$\tilde{z} \geq 0 \quad \checkmark \quad \sum_{U: e \in U} \tilde{z}_U \geq w_e \quad \forall e \quad - \text{maintained by integrality of } w_e \text{ even if } k=0.$$

However: $\sum \tilde{z}_U \lfloor \frac{1}{2} |U| \rfloor$ decreased: By laminarity and b/c U_i were inclusion-wise maximal $\Rightarrow$ disjoint. 

Conclusions :

1. $\exists$ integral dual solution with
Laminar support.
2. Primal is T.D.I.
3. Exercise: Use this result to obtain
the Tutte-Berge Thm.